Research and training report for the academic year 2016/2017

Merged mathematical models of the polluted atmosphere

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Presentation of fellow's background

- Master degree in Applied Mathematics, University of Szeged, Hungary
- Starting date of Marie Curie fellowship at the University of L'Aquila: 1st of October, 2016
- The first year at the University of L'Aquila has consisted of the following type of activities:
  - Training
  - Research

Training activities in the academic year 2016/2017

- **University of L'Aquila**: Numerical Analysis 2, High Performance Parallel Computing
- **Gran Sasso Science Institute**: Introduction to Finite Elements Method, Decay property for hyperbolic systems with dissipation, Advanced Topics on Numerical Methods, Introduction to the incompressible Navier-Stokes equation
- **Seminars, workshops**: PDEs Day (GSSI), School on Uncertainty Quantification for Hyperbolic Equations and Related Topics, CIME School 2017 - Mathematical Analysis of the Navier-Stokes Equations (Cetraro)

Research: Modeling of the atmosphere

The beginning step of this research project was to collect sets of equations that describe well the motion of the atmosphere, firstly without considering the effect of pollution. The nature / source of these equations are the following:

- conservation of momentum, Navier-Stokes equation
- conservation of mass (continuity equation)
- thermodynamics equation (conservation of energy)
- equation of state
- equation describing the quantity of water vapor (ocean analogy: salinity)

Different approaches for the equations

Using these equations we can create four main different versions of sets of equations that describe the atmosphere:

- Using the classical (x, y, z) cartesian coordinates,
- Using the spherical coordinates (θ, φ, z),
- Introducing a terrain-influenced coordinate η instead of z,
- Using the well-behaving, monotonous relationship between pressure and height and use p as a vertical coordinate.

An example set of equations

As an example, the equations of the atmosphere in (x, y, p) coordinates:

$$\frac{\partial \mathbf{v}}{\partial t} + \nabla_{\mathbf{v}}\mathbf{v} + \omega\frac{\partial \mathbf{v}}{\partial p} + 2\Omega\sin\theta\mathbf{k}\times\mathbf{v} + \nabla\Phi - L_{\mathbf{v}}\mathbf{v} = F_{\mathbf{v}}$$

$$\frac{\partial \Phi}{\partial p} + \frac{RT}{p} = 0, \nabla\cdot\mathbf{v} + \frac{\partial \omega}{\partial p} = 0$$

$$\frac{\partial T}{\partial t} + \nabla_{\mathbf{v}}T + \omega\frac{\partial T}{\partial p} - \frac{R\bar{T}}{c_p p}\omega - L_T T = F_T$$

$$\frac{\partial q}{\partial t} + \nabla_{\mathbf{v}}q + \omega\frac{\partial q}{\partial p} - L_q q = F_q$$

$$p = R\rho T$$

Equation describing pollution

The species continuity equation (equations of chemical kinetics) has the following form:

$$\frac{\partial C_j}{\partial t} + \nabla \cdot J_j = r_j,$$

where

- C_j is the local molar concentration of species j, (time variations in the species concentration),
- the components of J_j represent the local molar flux, (local motion of the species), and
- r_j is a volume based formation rate.

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Synchronising

To merge the equations of the atmosphere and the equation of pollution we use the following approach:

- we write the fluid equations of the mixture considered as a single fluid with denisty ρ_0, pressure p, temperature T, velocity $\mathbf{v}$,
- we consider the equations describing the evolution of each mass fraction (or species concentration) Y_i or (C_i).
- we add "connecting" / "glue" equations regarding the quantities of the species to sum up for the quantity presented in the atmosphere equation.

Synchronising

One of the issues about synchronising was the fact that the equations of the atmosphere and pollution are often not in the same coordinate system. For solving this we could either

- use Jacobians corresponding to the rescaled vertical coordinate in the equations, or
- use the Jacobian of the spherical coordinate transform, or
- simultaneously both of the above

We have chosen to deal only with the spherical coordinate change, since because of the quasi-horizontal nature of the atmosphere we will perform dimension reduction.

Current state part 1

$$\frac{\partial \mathbf{v}}{\partial t} + \nabla_{\mathbf{v}}\mathbf{v} + w\frac{\partial \mathbf{v}}{\partial z} + \frac{1}{\rho_0}\nabla p + 2\Omega \sin\theta \mathbf{k} \times \mathbf{v} - \mu_{\mathbf{v}}\Delta \mathbf{v} - \nu_{\mathbf{v}}\frac{\partial^2 \mathbf{v}}{\partial z^2} = 0$$

$$\frac{\partial p}{\partial z} = -\rho_0 g$$

$$\frac{\partial \rho_0}{\partial t} + \rho_0(\nabla \mathbf{v} + \frac{\partial w}{\partial z}) + \mathbf{v}\nabla\rho_0 + w\frac{\partial \rho_0}{\partial z} = 0$$

$$\frac{\partial T}{\partial t} + \nabla_{\mathbf{v}} T + w\frac{\partial T}{\partial z} - \mu_T \Delta T - \nu_T \frac{\partial^2 T}{\partial z^2} - \frac{RT}{p}\frac{\mathrm{d}p}{\mathrm{d}t} = Q_T$$

$$\frac{\partial q}{\partial t} + \nabla_{\mathbf{v}} q + w\frac{\partial q}{\partial z} - \mu_q \Delta q - \nu_q \frac{\partial^2 q}{\partial z^2} = 0$$

$$p = R\rho_0 T$$

Current state part 2

$$\frac{\partial \bar{\rho_i}}{\partial t} + \nabla_h \cdot (\bar{\rho_i} \bar{\mathbf{v}}) + \frac{\partial}{\partial z}(\bar{\rho_i} \bar{w}) - \bar{\rho} \nabla_h \cdot (K_{11}, K_{22}) \nabla \bar{Y}_i + \frac{\partial}{\partial z}(-\bar{\rho} K_{33} \frac{\partial \bar{Y}_i}{\partial z}) =$$

$$= R_{\rho_i}(\bar{\rho_1}, \ldots, \bar{\rho_N}) + Q_{\rho_i} + \frac{\partial(\bar{\rho_i})}{\partial t}|_{cld} + \frac{\partial(\bar{\rho_i})}{\partial t}|_{aero} + \frac{\partial(\bar{\rho_i})}{\partial t}|_{ping},$$

(which is a more refined version of the previous pollution equation)

$$\rho_0 = \sum_{i=1}^{N} \rho_i$$

Outlook

- Dimension reduction for the merged system. The 3D model is too complex to work with, on the other hand the phenomena can be seen as quasi-horizontal, which is why we perform dimension reduction for the merged system, the goal is to have a synchronized 2D system describing pollution and atmosphere.
- Solutions for the new, two-dimensional merged system.
- Numerical simulations of solutions.